

Akshansh Ranjan

+91 98350 14501 | akshanshranjan007@gmail.com | [Portfolio](#) | [LinkedIn](#) | [GitHub](#)

EDUCATION

BITS Pilani, Hyderabad Campus

2023 – 2027

B.E. Electrical and Electronics Engineering, Minor in Computer Science

CGPA: 8.02

Coursework: Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Database Systems, OOP, Linear Algebra, Probability & Statistics

TECHNICAL SKILLS

AI / ML: PyTorch, Scikit-learn, CNNs, XLM-Roberta, Hugging Face Transformers, NLP, RAG

LLM / Agents: LangGraph, LangChain, MCP, LlamaIndex, LangSmith, OpenAI/Gemini API, ChromaDB, BM25

Languages: Python, C++, SQL, JavaScript, Java

Backend / Data: FastAPI, PostgreSQL, Redis, Docker, Streamlit, Pandas, NumPy, Matplotlib

Tools: Git, GitHub, Jupyter, VS Code, Linux, Vercel

EXPERIENCE

ML Intern

May 2025 – Jul 2025

AI Gurukul

Remote

- Fine-tuned **XLM-Roberta** for sentence-level classification on code-mixed text across **5 Indian languages**.
- Achieved **91% weighted F1-score** on held-out test data across all language classification tasks.
- Built a token-level detector using **Gemini API + BM25**, covering **70+ languages** with 3x accuracy gain.
- Developed a **PDF RAG chatbot** with 500-token chunking, embedding retrieval, and multilingual prompting.
- Designed full **NLP pipeline**: preprocessing, tokenization, label balancing, and F1/confusion matrix eval.

PROJECTS

OrqFlow – Multi-Agent AI Platform | *LangGraph, FastMCP, FastAPI, Redis, PostgreSQL*

Jun 2026

- Built a **5-node LangGraph engine** (Supervisor, Researcher, Analyst, Coder, Extractor) with state persistence.
- Developed **3 FastMCP servers** for live SQL database querying, local filesystem manipulation, and web search.
- Engineered zero-downtime **LLM failover** switching between Groq and Gemini upon hitting HTTP 429 rate limits.
- Architected an **11-endpoint FastAPI backend** streaming real-time SSE tokens with Redis rate-limiting (10 req/min).
- Designed async **SQLAlchemy/Postgres schemas** storing chat sessions, user profiles, and extracted knowledge facts.

ShopSense – Hybrid Recommendation Engine | *Python, Sentence-Transformers, FastAPI, Redis, Vanilla JS, Vercel*

May 2026

- Built a **hybrid recommendation engine** combining ALS, BPR, and content-based filtering on **1M+ H&M transactions**.
- Trained **Sentence-Transformer** embeddings on **105K** product descriptions for user profile via sparse matrix projection.
- Implemented **MMR diversity reranking** to balance relevance and catalog coverage across **8 competing models**.
- Deployed **FastAPI** microservice featuring **Redis caching** with graceful in-memory degradation for **sub-50ms latency**.
- Built an offline **A/B evaluation** pipeline computing NDCG, MAP, and novelty across **3 user activity segments**.

Frequency Response Analysis of Power Transformers | *Python, TensorFlow, Xception, Streamlit*

Jul 2025 – Dec 2025

- Developed an automated **Xception deep learning system** replacing manual FRA interpretation with fault diagnostics.
- Curated a **synthetic dataset of 26,000+ samples** across 6 fault classes using signal processing augmentation.
- Achieved **98.4% test accuracy** with detailed per-class precision/recall and confusion matrix evaluation.
- Built a **Streamlit app** to visualize raw FRA signals, model predictions, and confidence scores for end users.
- Published and presented findings at **GCON–IEEE Conference, IIT Guwahati** (Jun 2026).

SyncNexus – Real-time Collaboration with RAG AI Assistant | *Socket.IO, ChromaDB, BullMQ, Redis*

Jun 2026

- Integrated **RAG-powered AI assistant** answering document queries with exact metadata citations via ChromaDB.
- Built resilient **document ingestion pipeline**: S3 upload → chunking → batch embedding → idempotent ChromaDB upserts.
- Used **Gemini text embeddings** for semantic vector storage and similarity-based chunk retrieval at query time.
- Implemented async LLM processing via **BullMQ** queues with exponential backoff, preventing WebSocket event loop blocking.
- Engineered secure **presigned S3 uploads** and stable keyset **cursor pagination** with Redis-backed tracking per room.

ACHIEVEMENTS

Published a research paper at IEEE GCON, IIT Guwahati introducing **Robust Difference Plot (RDP) Analysis**, a novel methodology for evaluating **Frequency Response Analysis (FRA)** plots to detect six different mechanical and electrical transformer faults.

Jun 2026

National Finalist (Top 2.5%) – Meta PyTorch OpenEnv Hackathon × Scaler: Built **BluffBuster** using **OpenEnv**, **Unsloth**, and **GRPO** to train language models via multi-style student simulators and calibrated belief tracking. Selected among **800 teams from 31,000+ registrations**.

Apr 2026

Sports & Communication: Represented school and university **football** teams in competitive tournaments. Fluent in English and Hindi.